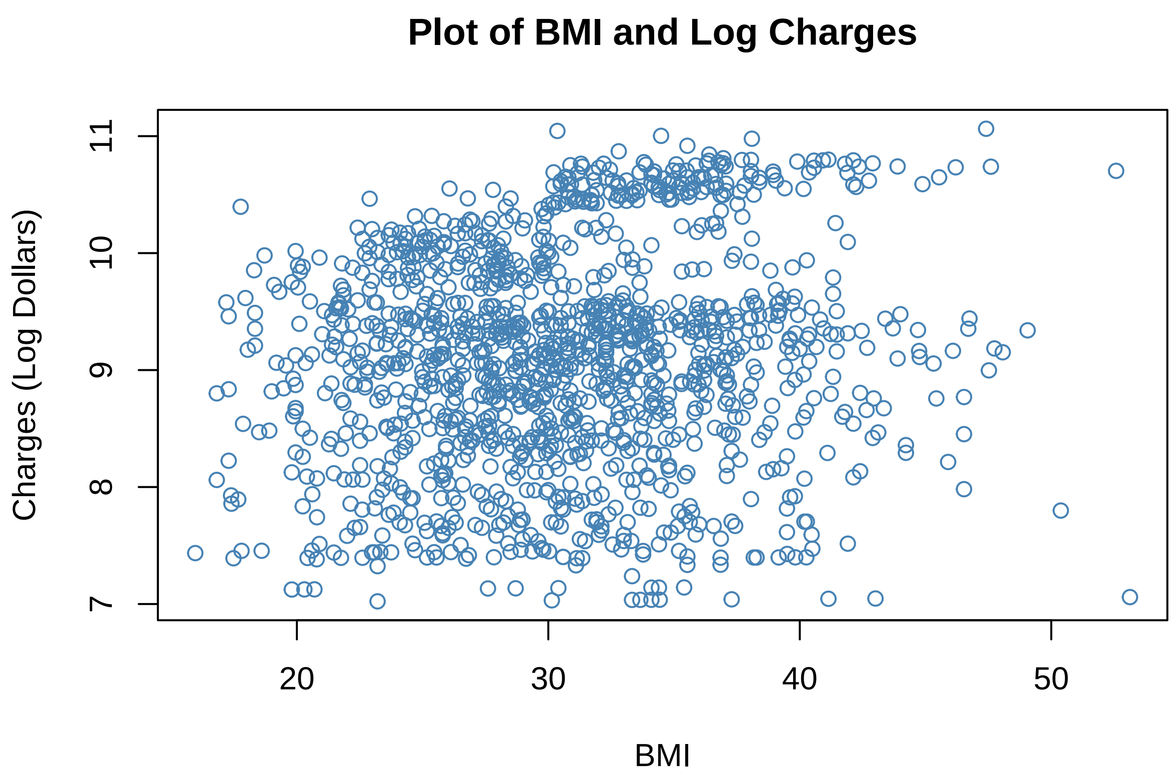
Simple Linear Model Code

```r
# Sets seed for reproducibility
set.seed(42)
```

```r
# Reads in data frame
df <- read.csv("insurance.csv")

# Creates log transformation of charges
df$log_charges <- log(df$charges)

# Sets categorical variables as factors
df$sex <- as.factor(df$sex)
df$children <- as.factor(df$children)
df$region <- as.factor(df$region)
df$smoker <- as.factor(df$smoker)

# Creates training and testing sets
n <- nrow(df)
train_index <- sample(1:n, size = 0.7 * n)

df_train <- df[train_index, ]
df_test  <- df[-train_index, ]

# Prints snipit of data frame
head(df)
```

```
##   age    sex    bmi children smoker    region   charges log_charges
## 1  19 female 27.900        0    yes southwest 16884.924    9.734176
## 2  18   male 33.770        1     no southeast  1725.552    7.453302
## 3  28   male 33.000        3     no southeast  4449.462    8.400538
## 4  33   male 22.705        0     no northwest 21984.471    9.998092
## 5  32   male 28.880        0     no northwest  3866.855    8.260197
## 6  31 female 25.740        0     no southeast  3756.622    8.231275
```

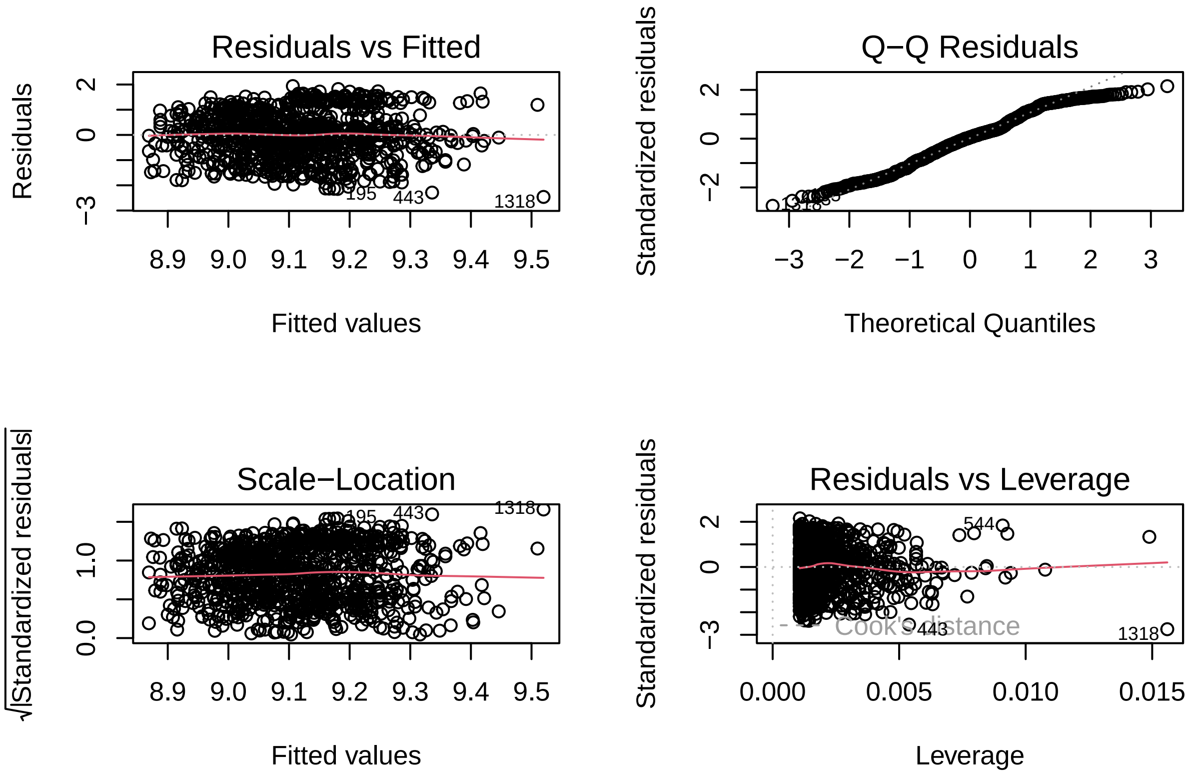
```r
# Plots bmi against log charges
plot(df$bmi, df$log_charges, main="Plot of BMI and Log Charges", xlab="BMI",
     ylab="Charges (Log Dollars)", col="steelblue")
```



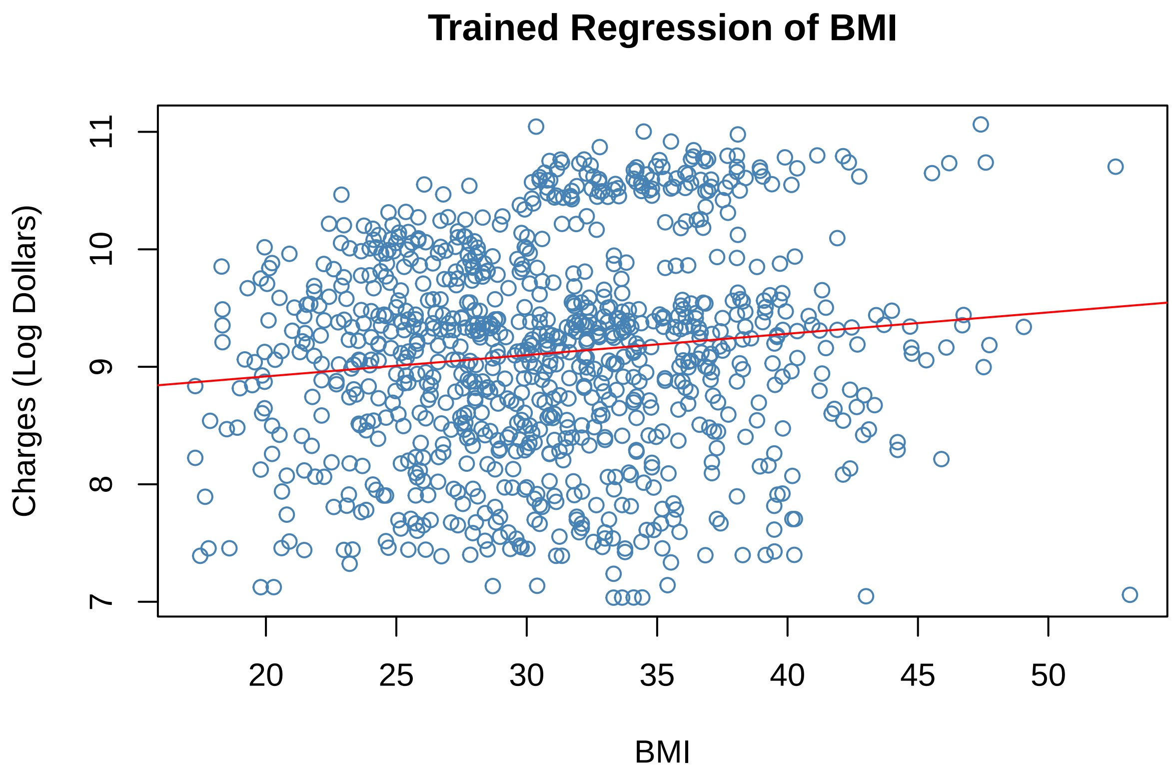

```r
# Creates regression model of just bmi
lm1 <- lm(log_charges ~ bmi, data=df_train)
# Prints summary of model
summary(lm1)
```

```
##
## Call:
## lm(formula = log_charges ~ bmi, data = df_train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.46066 -0.62192  0.05065  0.65695  1.93815
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 8.554862   0.153418  55.762  < 2e-16 ***
## bmi         0.018162   0.004878   3.723 0.000208 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9011 on 934 degrees of freedom
## Multiple R-squared:  0.01462,    Adjusted R-squared:  0.01357
## F-statistic: 13.86 on 1 and 934 DF,  p-value: 0.0002085
```

```
# Prints diagnostic plots for model
par(mfrow=c(2,2))
plot(lm1)
```



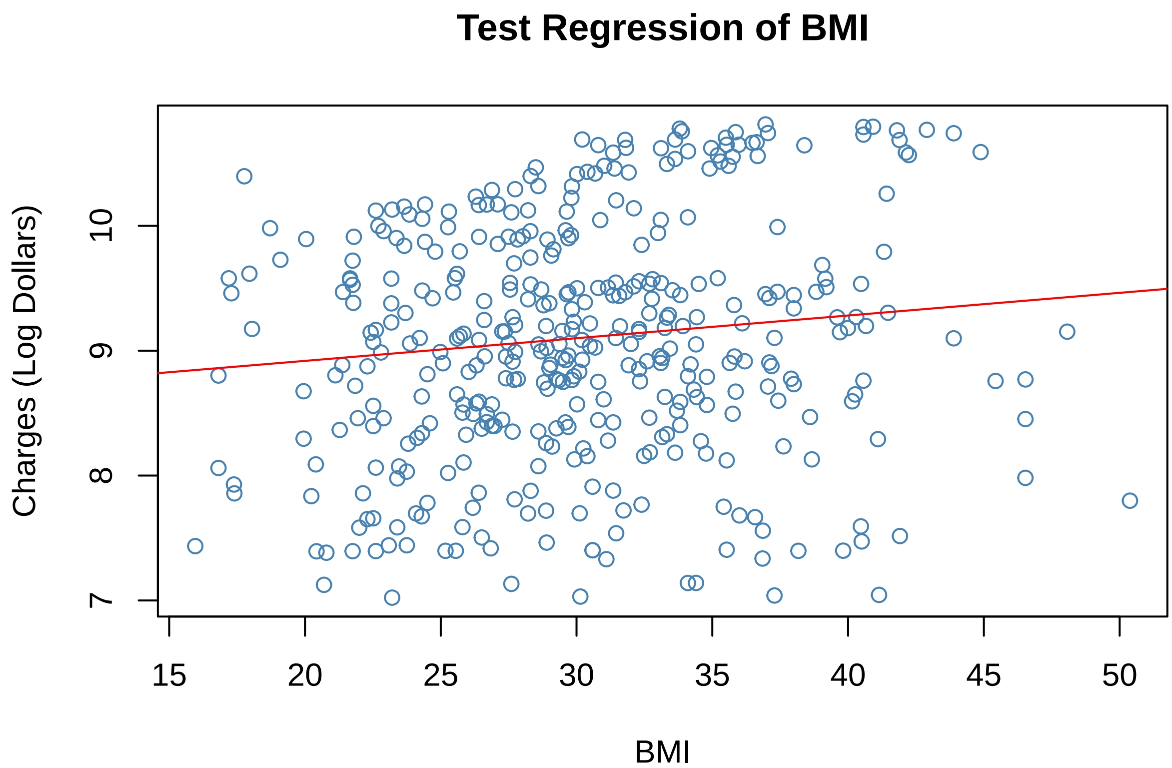

```
# Plots trained regression line on training data
plot(df_train$bmi, df_train$log_charges, main="Trained Regression of BMI",
     xlab="BMI", ylab="Charges (Log Dollars)", col="steelblue")
abline(lm1, col="red")
```




```r
# Calculates rmse for training set
sqrt(mean((df[as.numeric(names(lm1$fitted.values)),]$charges - exp(lm1$fitted.values))^2))
```

```
## [1] 12677.61
```

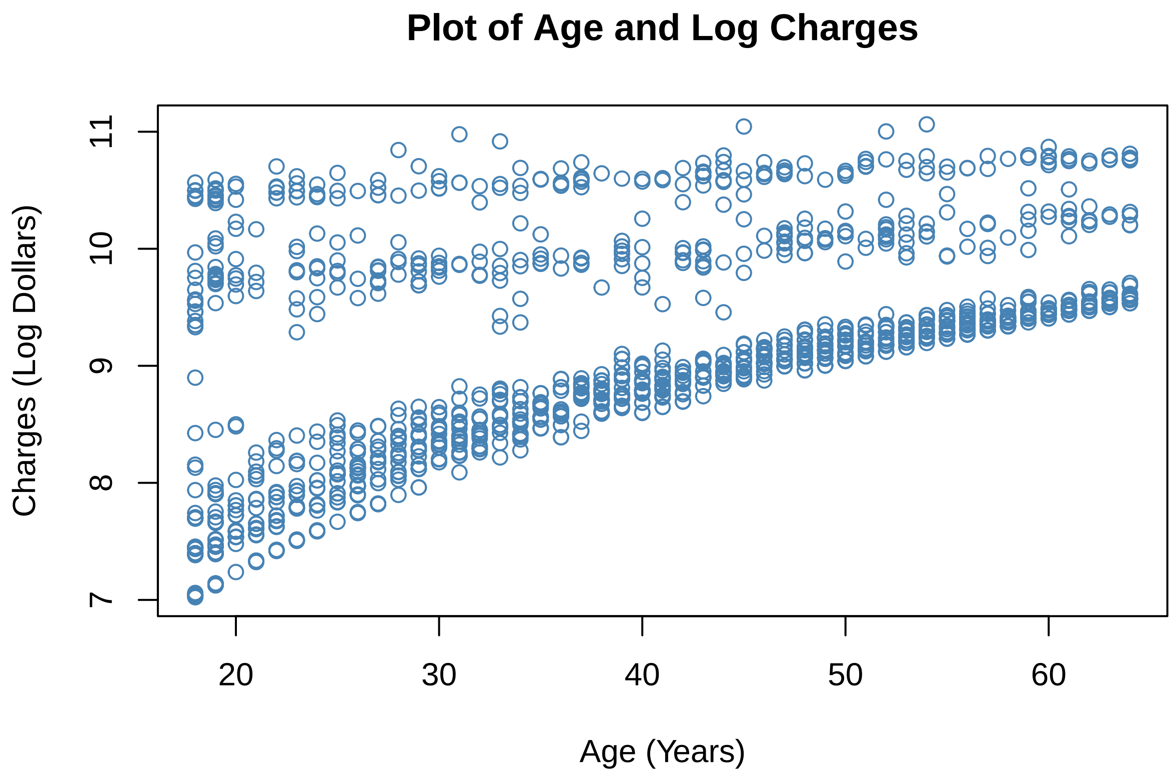
```r
# Plots trained regression line on test set
plot(df_test$bmi, df_test$log_charges, main="Test Regression of BMI",
     xlab="BMI", ylab="Charges (Log Dollars)", col="steelblue")
abline(lm1, col="red")
```




```r
# Calculates rmse for test set
pred <- exp(predict(lm1, newdata=df_test))

sqrt(mean((df_test$charges - pred)^2))
```

```
## [1] 12579.86
```

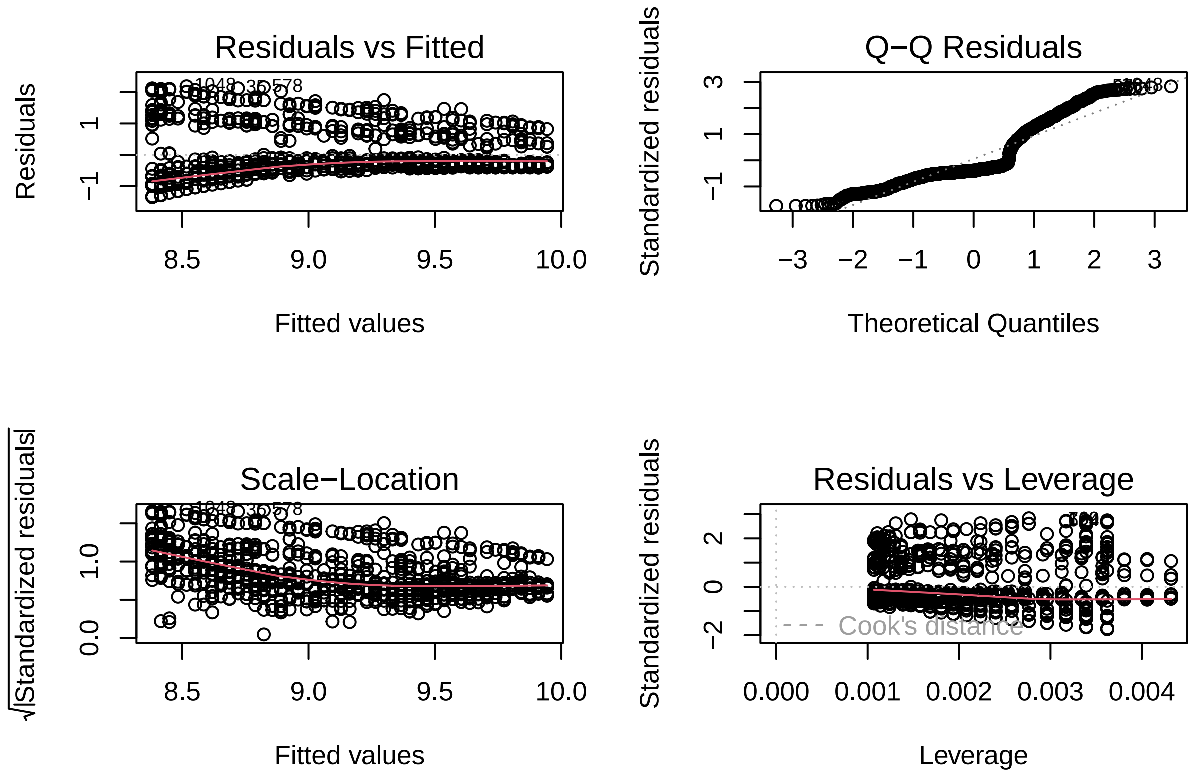
```r
# Plots age against log charges
plot(df$age, df$log_charges, main="Plot of Age and Log Charges",
     xlab="Age (Years)", ylab="Charges (Log Dollars)", col="steelblue")
```




```r
# Creates regression model of just age
lm2 <- lm(log_charges ~ age, data=df_train)
# Prints model summary
summary(lm2)
```

```
##
## Call:
## lm(formula = log_charges ~ age, data = df_train)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.3463 -0.4155 -0.3048  0.5025  2.1859
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 7.770247   0.075947  102.31   <2e-16 ***
## age         0.033959   0.001808   18.78   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7733 on 934 degrees of freedom
## Multiple R-squared:  0.2742, Adjusted R-squared:  0.2734
## F-statistic: 352.8 on 1 and 934 DF,  p-value: < 2.2e-16
```

```r
# Prints diagnostic plots for model
par(mfrow=c(2,2))
plot(lm2)
```




```r
# Plots trained regression line on training set
plot(df_train$age, df_train$log_charges, main="Trained Regression of Age",
     xlab="Age (Years)", ylab="Charges (Log Dollars)", col="steelblue")
abline(lm2, col="red")
```


```r
# Computes rmse for training set
sqrt(mean((df[as.numeric(names(lm2$fitted.values)),]$charges - exp(lm2$fitted.values))^2))
```

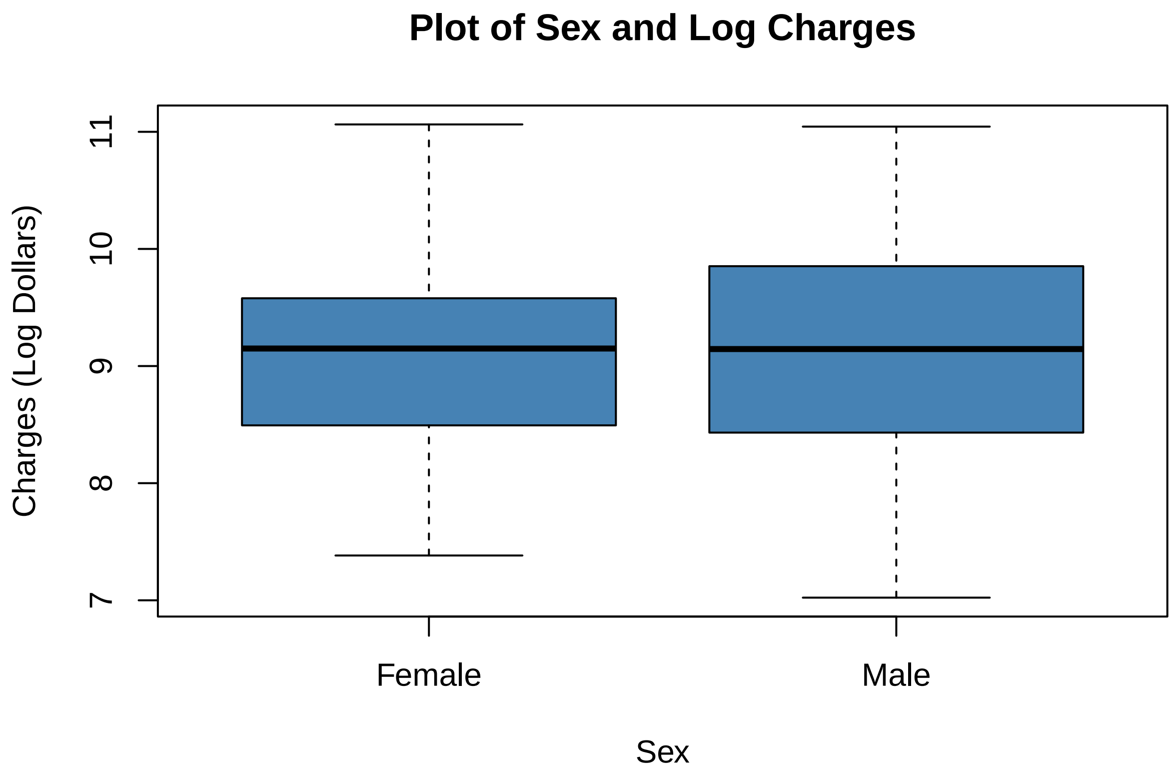
```
## [1] 12062.52
```

```r
# Plots trained regression line on test set
plot(df_test$age, df_test$log_charges, main="Test Regression of Age",
     xlab="Age (Years)", ylab="Charges (Log Dollars)", col="steelblue")
abline(lm2, col="red")
```


```
# Calculates rmse for the test set
pred <- exp(predict(lm2, newdata=df_test))

sqrt(mean((df_test$charges - pred)^2))
```

```
## [1] 11993.45
```

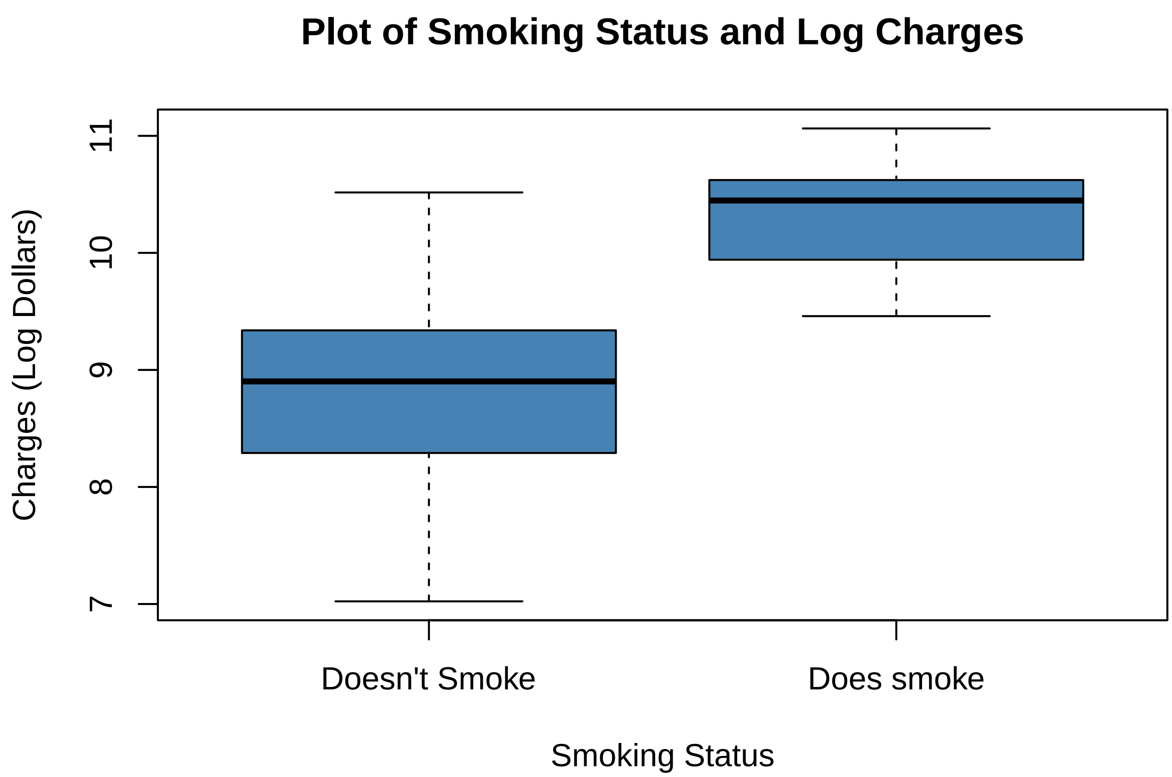
```
# Creates box plots of charges for males and females
plot(df$sex, df$log_charges, main="Plot of Sex and Log Charges", xlab="Sex",
     ylab="Charges (Log Dollars)", col="steelblue", xaxt="n")
axis(1, at = 1:2, labels = c("Female", "Male"))
```




```r
# Creates model of each category in sex
lm3 <- lm(log_charges ~ sex, data=df)
# Prints summary of model
summary(lm3)
```

```
##
## Call:
## lm(formula = log_charges ~ sex, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.08103 -0.63503  0.04289  0.62600  1.96962
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  9.09343    0.03575 254.353   <2e-16 ***
## sexmale      0.01035    0.05030   0.206    0.837
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9199 on 1336 degrees of freedom
## Multiple R-squared:  3.172e-05,  Adjusted R-squared:  -0.0007168
## F-statistic: 0.04238 on 1 and 1336 DF,  p-value: 0.8369
```

```r
# Box plots of smoking status agaist log charges
plot(df$smoker, df$log_charges, main="Plot of Smoking Status and Log Charges",
     xlab="Smoking Status", ylab="Charges (Log Dollars)", col="steelblue",
     xaxt="n")
axis(1, at = 1:2, labels = c("Doesn't Smoke", "Does smoke"))
```

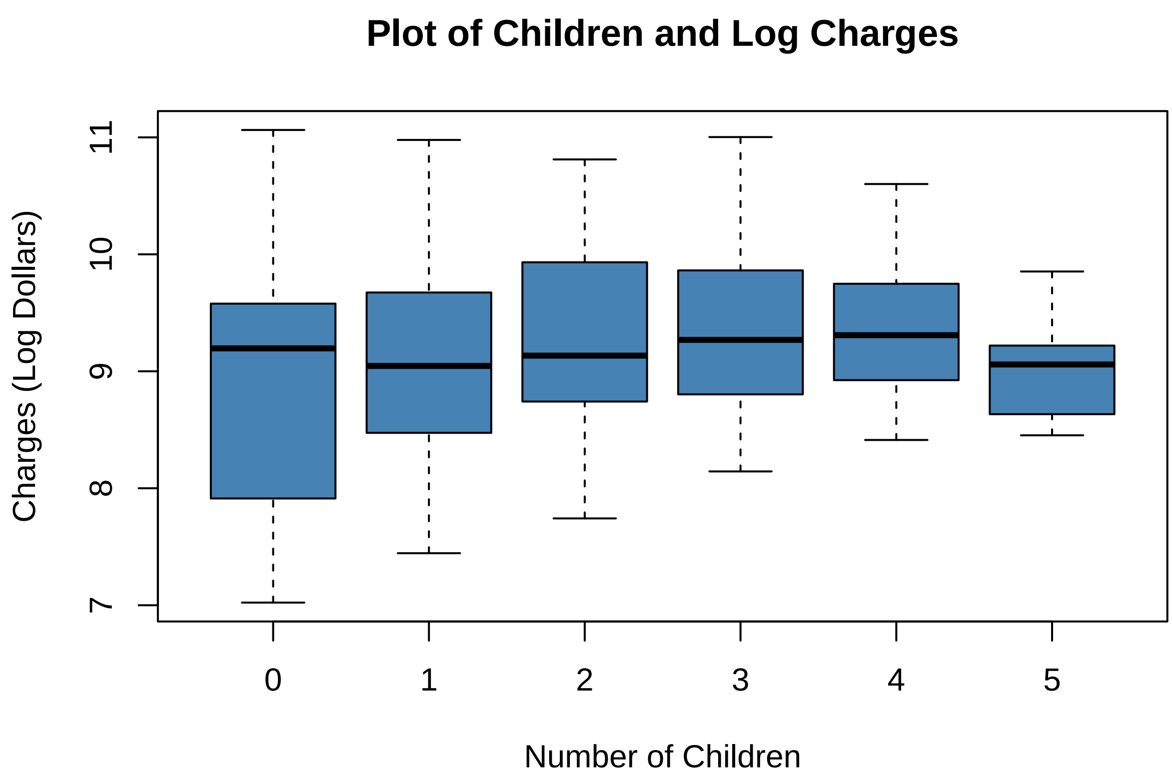



```r
# Creates model of each status in smoker
lm4 <- lm(log_charges ~ smoker, data=df)
# Prints summary of model
summary(lm4)
```

```
##
## Call:
## lm(formula = log_charges ~ smoker, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.7655 -0.4370  0.1237  0.4785  1.7280
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8.78823    0.02105  417.52   <2e-16 ***
## smokeryes    1.51588    0.04651   32.59   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 0.6866 on 1336 degrees of freedom
## Multiple R-squared:  0.4429, Adjusted R-squared:  0.4425
## F-statistic:  1062 on 1 and 1336 DF,  p-value: < 2.2e-16
```

```r
# Plots number of children against log charges
plot(df$children, df$log_charges, main="Plot of Children and Log Charges",
     xlab="Number of Children", ylab="Charges (Log Dollars)", col="steelblue")
```

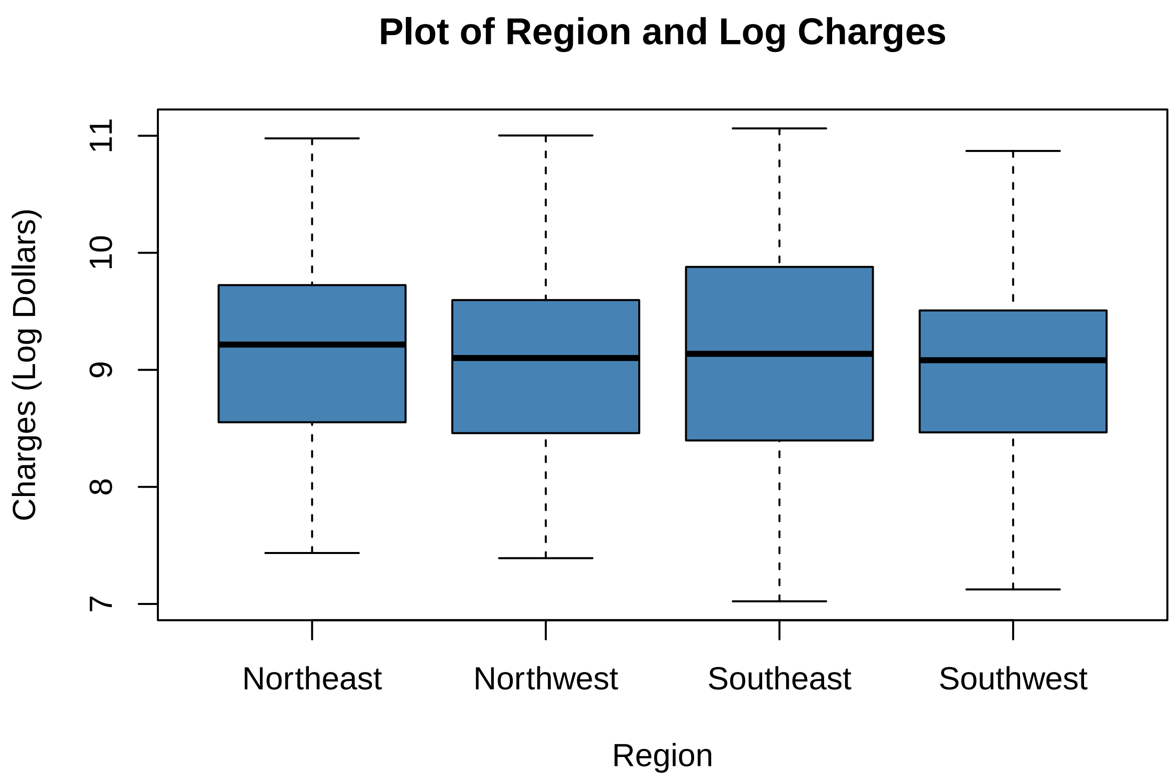



```r
# Creates regression model number of children
lm5 <- lm(log_charges ~ children, data=df)
# Prints model summary
summary(lm5)
```

```
##
## Call:
## lm(formula = log_charges ~ children, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.91242 -0.67963  0.01206  0.61258  2.12787
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8.93517    0.03780 236.403  < 2e-16 ***
```

```
## children1     0.15964    0.06292   2.537   0.0113 *
## children2     0.36309    0.06961   5.216 2.12e-07 ***
## children3     0.43374    0.08156   5.318 1.23e-07 ***
## children4     0.42384    0.18501   2.291   0.0221 *
## children5     0.06592    0.21676   0.304   0.7611
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9055 on 1332 degrees of freedom
## Multiple R-squared:  0.03383,    Adjusted R-squared:  0.0302
## F-statistic: 9.327 on 5 and 1332 DF,  p-value: 9.566e-09
```

```r
# Plots regions agains log charges
plot(df$region, df$log_charges, main="Plot of Region and Log Charges",
     xlab="Region", ylab="Charges (Log Dollars)", col="steelblue", xaxt="n")
axis(1, at = 1:4,
     labels = c("Northeast", "Northwest", "Southeast", "Southwest"))
```




```r
# Creates regression model of regions
lm6 <- lm(log_charges ~ region, data=df)
# Prints model summary
summary(lm6)
```

```
##
## Call:
```

```
## lm(formula = log_charges ~ region, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.09965 -0.64192  0.03622  0.62333  1.94064
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)      9.16877    0.05106 179.562   <2e-16 ***
## regionnorthwest -0.09903    0.07216  -1.372   0.1701
## regionsoutheast -0.04637    0.07020  -0.660   0.5091
## regionsouthwest -0.13767    0.07216  -1.908   0.0566 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9191 on 1334 degrees of freedom
## Multiple R-squared:  0.003143,   Adjusted R-squared:  0.0009012
## F-statistic: 1.402 on 3 and 1334 DF,  p-value: 0.2406
```